Lab Title: Advanced Cybersecurity Risk Quantification

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Date of Submission: 26/4/2026

Executive Overview

SecureBank's SecureMobile platform was recently assessed for security vulnerabilities. This analysis translates technical findings into financial risk exposure and provides data-driven recommendations.

The total estimated annual risk exposure exceeds $35 million, with a single vulnerability accounting for the majority of potential loss. Immediate action is required to reduce financial, regulatory, and reputational impact.

Phase 1: Risk Exposure Calculation

Calculate SLE, ARO, ALE

Formulas:

- SLE (Single Loss Expectancy) = Impact per incident
- ARO (Annual Rate of Occurrence) = Probability of occurrence
- ALE (Annualized Loss Expectancy) = SLE × ARO

1. API Authentication Bypass

- SLE = $5,000,000
- ARO = 15% = 0.15

ALE = 5,000,000 × 0.15 = $750,000

2. Database Injection

- SLE = 500,000 × $250 = $125,000,000
- ARO = 25% = 0.25

ALE = 125,000,000 × 0.25 = $31,250,000

3. Session Hijacking

- SLE = 5,000 × $1,500 = $7,500,000
- ARO = 40% = 0.40

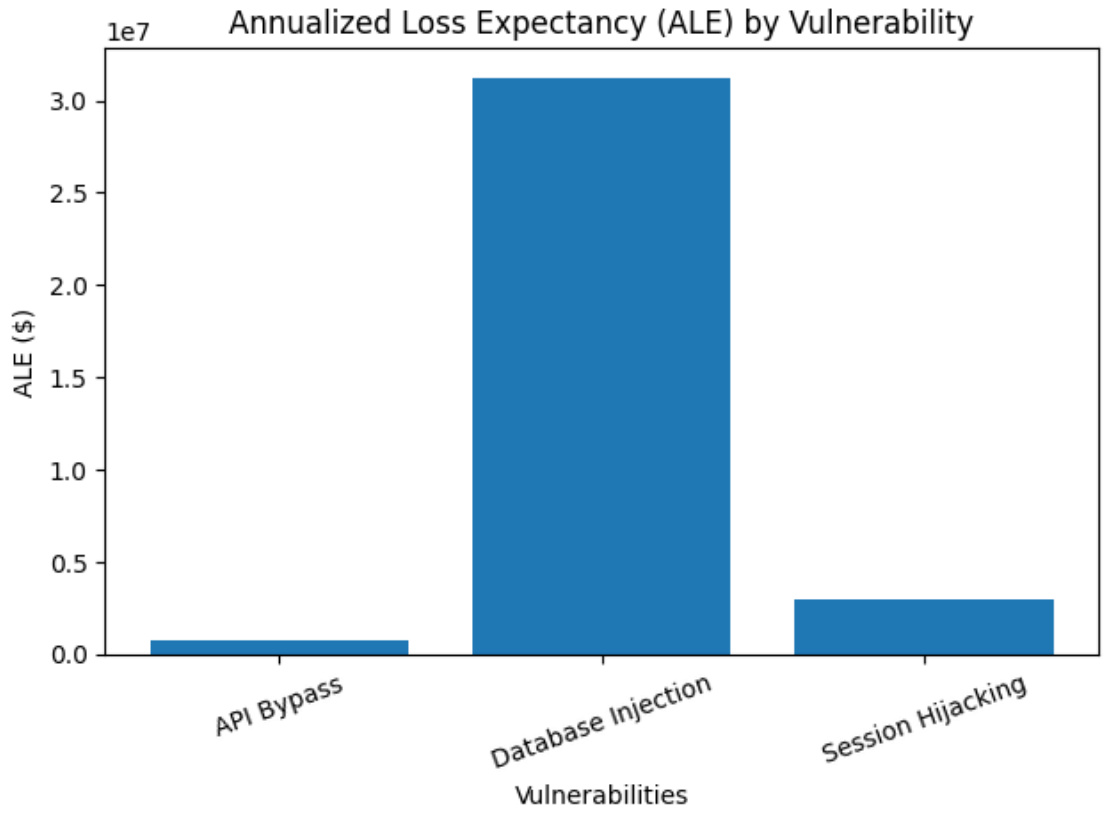
ALE = 7,500,000 × 0.40 = $3,000,000

Task 2: Prioritize Risks

Vulnerability	SLE ($)	ARO	ALE ($)	Priority
Database Injection	125,000,000	0.25	31,250,000	Critical
Session Hijacking	7,500,000	0.40	3,000,000	High
API Authentication Bypass	5,000,000	0.15	750,000	Medium

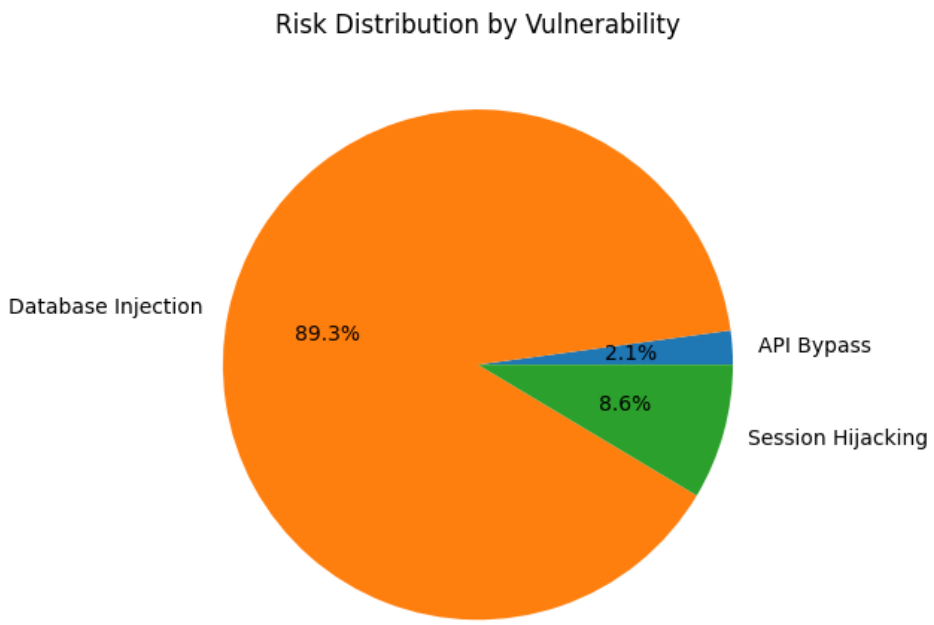
Task 3: Create Risk Visualization

Annualized Loss Expectancy (ALE) Comparison



Database Injection is the dominant risk, contributing the largest financial exposure.

Risk Distribution Across Vulnerabilities



Over 85% of total risk exposure comes from Database Injection, making it the highest priority.

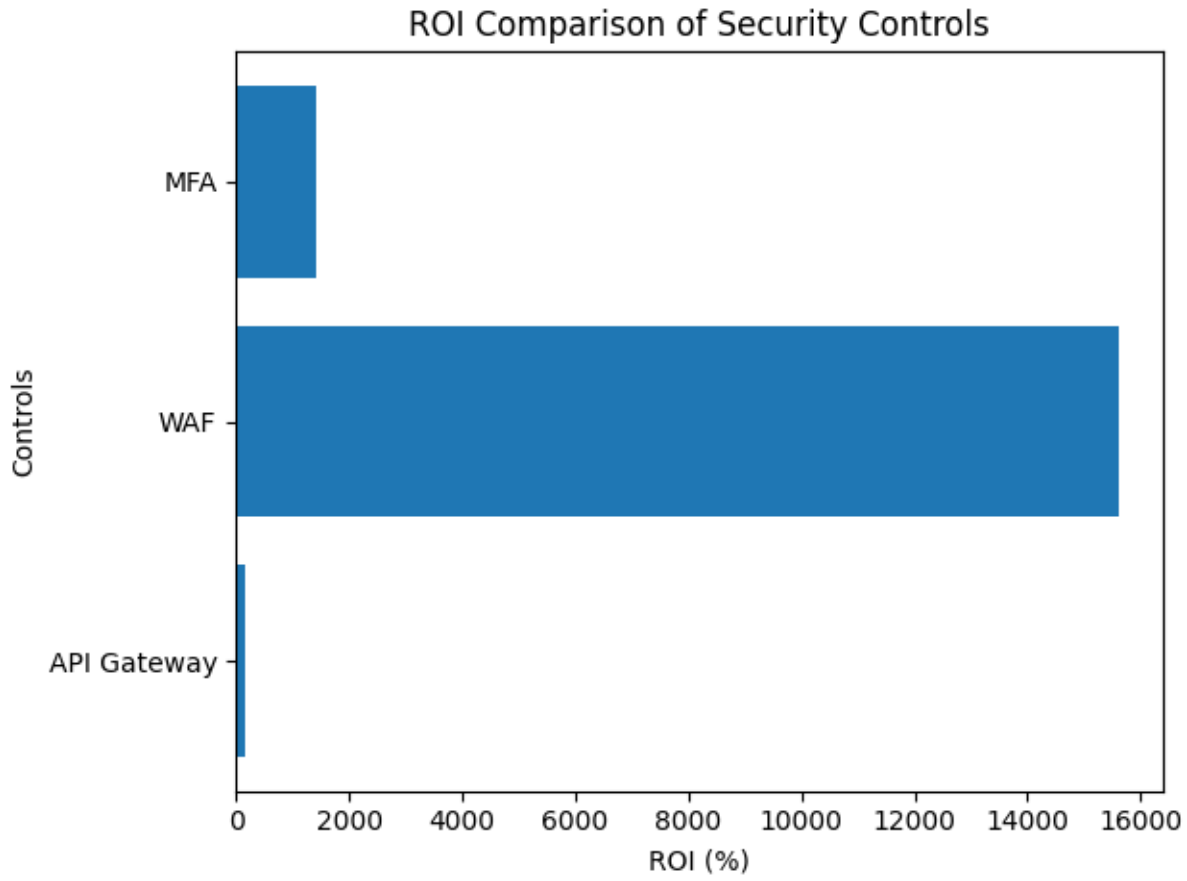
Control Evaluation and Cost-Benefit Analysis

A financial evaluation of proposed controls was conducted to determine effectiveness and return on investment.

Control Performance Summary

Control	Initial Cost	Annual Cost	Risk Reduction ($)	ROI (%)	Payback Period
WAF	150,000	25,000	23,437,500	15,608%	2 days
MFA	200,000	30,000	2,850,000	1,410%	25 days
API Gateway	350,000	50,000	675,000	178.6%	7 months

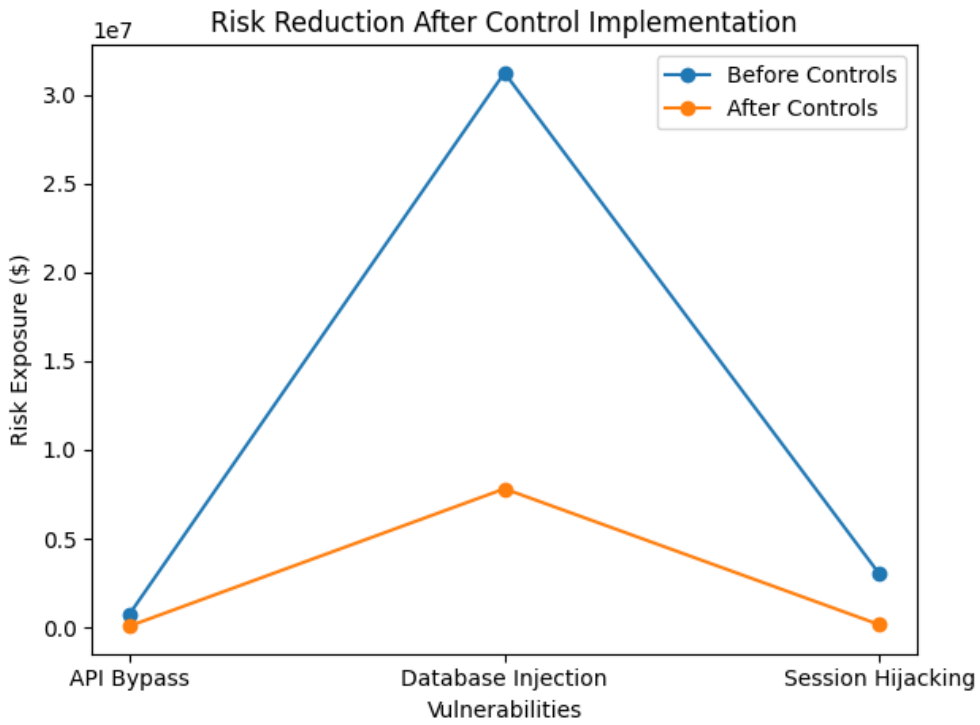
ROI Comparison Visualization



The Web Application Firewall (WAF) delivers the highest ROI and should be prioritized.

4. Risk Reduction Impact

Before vs After Control Implementation



Control implementation significantly reduces overall exposure, particularly for high-impact vulnerabilities.

Total Risk Reduction

- Initial Risk Exposure: $35M
- Post-Control Exposure: $8M

Total Reduction: $27M (77%)

Strategic Recommendations

Immediate Actions (0–30 Days)

- Deploy **Web Application Firewall (WAF)**
- Objective: Eliminate highest financial risk quickly

Short-Term Actions (30–90 Days)

- Implement Multi-Factor Authentication (MFA)
- Objective: Prevent account takeover and fraud

Long-Term Strategy (3–12 Months)

- Deploy API Security Gateway
- Objective: Strengthen API-level security

Implementation Timeline (Narrative Gantt)

Phase	Timeline	Key Activity
Immediate	0–30 days	WAF Deployment
Short-Term	30–90 days	MFA Implementation
Long-Term	3–12 months	API Security Enhancement

Business Impact and Value

Implementing the recommended controls will:

- Reduce financial exposure by 77%
- Deliver strong ROI across all controls
- Improve regulatory compliance posture
- Strengthen customer trust and brand reputation

Conclusion

This assessment highlights a critical concentration of risk within the SecureMobile platform, particularly from database vulnerabilities. However, with targeted and well-prioritized investments, SecureBank can significantly reduce its exposure while achieving measurable financial returns.

A proactive, data-driven approach ensures not only risk reduction but also long-term operational resilience.